

Md Abdullah-Al Mamun

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CAREER PROFILE

Dedicated Researcher & Data Scientist with a solid foundation in data analysis and machine learning, supported by a master's degree in data science from Tampere University. With over years of professional experience, I have honed my skills across various fields, including programming, databases, Android development, version control, cloud computing, test automation, and laboratory and industrial techniques. I possess specialized skills in machine learning algorithms, image analysis, and natural language processing (NLP), demonstrating my ability to apply data-driven solutions to real-world problems. My diverse background, including web development and art as a hobby, enables me to bring a creative approach to data science.

ACADEMIC AND PROFESSIONAL SKILLS

- **Programming:** Python, JavaScript, R (statistical analysis), MATLAB (coursework and scientific visualization), Java (basic, Android), TypeScript (basic)
- **Databases:** SQL, PostgreSQL, MySQL.
- **Visualization:** Matplotlib, Seaborn, Plotly, ggplot, Microsoft Power BI.
- **Machine Learning Skills:** NumPy, Pandas, SciPy, Scikit-Learn, NLP, Text Data Analysis, Image Analysis.
- **Deep Learning:** TensorFlow, PyTorch, Neural Networks.
- **Generative AI & LLM:** Large Language Model Integration, LLM-based Data Extraction, GenAI Applications, Prompt Engineering.
- **Development & Version Control:** Android App Development (Java), Git, GitHub.
- **Cloud Platforms:** AWS, Microsoft Azure.
- **Testing:** Unit Testing, Integration Testing.
- **Laboratory and industrial skills:** Excel, SAP, FidaWare, Power BI, Analytic suite, 3D Scanning, Digital microscopy.
- **Research domains:** Environmental Science, ML & DL Applications, Artificial Intelligence, Statistical Modeling.
- **Spatial Analysis:** GIS Mapping (QGIS, ArcGIS), Geostatistical Analysis, Spatial Modeling.

KEY PROJECTS

LLM & GenAI projects

September 2025

Website: <https://graham.fi/>

Project details: [Graham Defensive Investor Analyzer](#)

Developed a full-stack web application that extracts key financial metrics from PDF annual reports using **LLM-based** extraction, rule-based parsing and web scraping, then evaluates them against Benjamin Graham's seven defensive investor criteria. Enabled automated pass/fail investment recommendations by integrating a **FastAPI backend**, **NLP pipelines**, a data integration layer and an interactive frontend with a manual calculator.

Machine Learning, and Deep Learning Projects

January 2020 – Present

- Explored **NLP** techniques using Python and libraries such as NLTK, spaCy, and scikit-learn. Conducted tasks including text preprocessing, tokenization, and sentiment analysis. Trained and evaluated machine learning models for text classification, achieving an impressive **92% accuracy** on the test dataset.
- Implemented a convolutional neural network (**CNN**) using PyTorch and PyTorch Lightning. Trained the model on a balanced emotion dataset and achieved a commendable accuracy of **79.29%**. Employed data augmentation techniques and handled class imbalance to improve the model's performance on facial emotion recognition.
- Developed and successfully completed **15+ highly demanding machine learning projects**, showcasing expertise in classification and regression problems. **Classification projects** include Health Dataset - Diabetes, Predicting Future Transactions - Banking Dataset, Multiple Sclerosis Disease and so on. **Regression projects** include Credit Risk Classification, Wine Quality, Sleep Efficiency Dataset, Waiter's Tips, Flight Delay and so on.

- Developed **50+ Python projects**, showcasing expertise in Advanced Python and utilizing various libraries.

R programming projects**October 2022**

- Analysed the relationship between corruption and development using data from the United Nations Human Development Report 2018.
- Cleaned and merged the data, and utilized R programming language with packages like ggplot2, plotly, and shiny to create interactive and visually appealing data visualizations.
- Compared the Corruption Perceptions Index (CPI) and the Human Development Index (HDI) to illustrate the correlation between corruption and development.

Android Applications**December 2017****Know Finnish:** Finnish language Learning App

- Created and published a Java-based app on Google Play Store as an independent Android developer.
- The app aims to help foreigners learn the Finnish language and has garnered over 1000 downloads.

Business Intelligence and Data visualization projects**January 2021-Present**

- Developed the "Personal Finance Analysis Dashboard" using Power BI to visualize transaction data.
- Utilized advanced **DAX** calculations, graphing techniques, and **dashboard design** principles to create an interactive and informative dashboard.
- Cleaned and organized the data to gain insights into spending patterns by location, date, and vendor, highlighting areas of higher expenditure and types of products purchased.
- Developed comprehensive visualizations using **Power BI** to explore the spread of the virus, mortality rates, and the effectiveness of government interventions.
- Collaborated with a team of data enthusiasts to analyze and present findings, aiming to raise awareness about the ongoing global crisis.

PEER-REVIEWED PUBLICATIONS : (10)**Google Scholar Profile****March 2024 – Present**

1. Islam, A. R. M. T., **Mamun, M. A.**, Uddin, M. N., Sowrav, S. F. F., Alam, M. N. E., Khan, S. R., Chowdhury, M. H., & Choudhury, T. R. (2026). *Explainable and physics-informed machine learning for seasonal water quality prediction in the monsoon-driven Padma River Basin, Bangladesh*. *Environmental Geochemistry and Health*, 48, Article 140. <https://doi.org/10.1007/s10653-026-03031-z>
2. **Mamun, M. A.**, Aktar, M. N., Uddin, M. N., Chowdhury, M. H., Islam, M. S., Rahman, M. S., Uddin, M. R., Hossain, M. J., Zahid, A., Senapathi, V., & Islam, A. R. M. T. (2026). *Hybrid data-driven framework for interpretable prediction of nitrate and sulfate risks in coastal aquifers*. *Science of the Total Environment*, 1011, 181190. <https://doi.org/10.1016/j.scitotenv.2025.181190>
3. Zesha, J. H., **Mamun, M. A.**, Islam, A. R. M. T., Habib, M. A., & Khan, R. (2026). *First probabilistic radiological risk appraisal of Bay of Bengal beach sands: Explicit spatial hot-spot analysis*. *Marine Pollution Bulletin*, 223, 118966. <https://doi.org/10.1016/j.marpolbul.2025.118966>
4. Islam, A. R. M. T., Varol, M., Mallick, J., **Mamun, M. A.**, Mia, M. Y., Siddique, M. A. B., Islam, M. S., & Aktar, M. N. (2026). *Tracing source footprints of heavy metal(oid)s in coastal soils using traditional statistical techniques and machine learning data-driven models*. *Marine Pollution Bulletin*, 222(Part 1), 118701. <https://doi.org/10.1016/j.marpolbul.2025.118701>
5. Islam, A. R. M. T., **Mamun, M. A.**, Hasan, M., Aktar, M. N., Uddin, M. N., Siddique, M. A. B., Chowdhury, M. H., Islam, M. S., Bari, A. B. M. M., Idris, A. M., & Senapathi, V. (2025). *Optimizing coastal groundwater quality predictions: A novel data mining framework with cross-validation, bootstrapping, and entropy analysis*. *Journal of Contaminant Hydrology*, 269, 104480. <https://doi.org/10.1016/j.jconhyd.2024.104480>

WORK EXPERIENCE (5+ years)

Master's Thesis worker

Valmet Automotive EV power Oy

Salo, Finland

March 2022 – June 2023

- Thesis Topic: “Map out the best BI tool and SPC software: selection and rationalization process.” [Trepo](#)
- Utilized Python, Power BI, and Excel to analyze and visualize data for evaluating and selecting the optimal Business Intelligence (BI) tool and Statistical Process Control (SPC) software.
- Recommended Microsoft Power BI based on comprehensive analysis, leading to its successful adoption by Valmet as the preferred BI tool.

Laboratory and quality Operator

Valmet Automotive EV power Oy

Salo, Finland

August 2021 – November 2023

- Conducted product audits and quality checks for 48-volt battery packs and Y7 batteries for Volvo automobiles, ensuring adherence to company standards and maintaining high-quality products.
- Achieved a remarkable quality rate of over 98%, with only 1-2% of products not meeting the required standards. Led investigations in the laboratory to identify the root causes of quality issues, taking appropriate actions to address them and prevent their recurrence.
- Performed various routine laboratory tasks, including sample collection and preparation, experimental procedures, and result analysis.
- Collected and analyzed data on a daily basis from the battery production line using Excel, focusing on parameters such as defect rates, inspection results, yield rates, and failure analysis.
- Visualized with Power BI and present data findings, contributing to informed decision-making processes. Maintained accurate and up-to-date records of quality control metrics in the company database, providing valuable insights for future reference and continuous improvement.

Web Developer

GOInternational Finland - GOI Finland Oy

Helsinki, Finland

July 2016 – Dec 2016

- Developed websites using HTML, CSS, and WordPress to promote the Finnish education system to international markets, specifically targeting countries like Nepal and Sri Lanka.
- Ensured the websites remained up to date by optimizing content for search engines (SEO).

Web and Media Designer

Helsinki Metropolia University of Applied Sciences (2015).

EDUCATION

- **Master's** degree in **computing science** with focus on **Data Science** at Tampere University, 2020 – 2024.
- **Bachelor's** degree in **information technology** (Major in **Software Engineering**) at Metropolia University of Applied Sciences, 2013 – 2017.
- **B.Sc. in Biotechnology and Genetic Engineering**, Mawlana Bhashani Science and Technology University, Bangladesh, 2006 – 2010.

REFERENCES

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